

1. ER Models (1 point)

- a. Notation: Standard ER symbols and cardinality constraints (e.g., 1:1, 1:N, M:N).
- b. Transformation: Mapping ER diagrams to relational schemas; ensuring schema normalization.

2. Query Languages (2 points)

- a. Formalisms: Relational Algebra (RA) and Tuple Relational Calculus (TRC).
- b. SQL: Writing optimized, concise queries; understanding functional restrictions (e.g., aggregation functions).

3. Database Constraints (1 point)

- a. Formal Definition: Translating business rules into RA or TRC expressions.
- b. Implementation: SQL constraints (CHECK, ASSERTIONS, etc.) and Procedural logic (can use pseudo-code).

4. Serialization & Concurrency (2 points)

- a. Theory: Analyzing schedule serializability.
- b. Protocols: Two-Phase Locking (2PL), Deadlock detection and resolution.
- c. Properties: Recoverable and Cascadeless schedules.

5. Recovery (1 point)

- a. Mechanisms: Redo-only, Undo-only, and Undo/Redo protocols.
- b. Failure Handling: Principles of system recovery.

6. Functional Dependencies (FD) & Normalization (1 point)

- a. FD Analysis: Computing closures and candidate keys.
- b. Normalization: Identifying Normal Forms (NF) and ensuring lossless-join properties.

7. Concurrency Control Execution (1 point)

- a. Transaction Schedules: Analyzing execution interleaving and listing potential outcomes or anomalies.

8. Query Optimization (1 point)

- a. Cost Estimation: Calculating I/O costs.
- b. Physical Access: Evaluating performance impacts of B-Tree indexing vs. Hashing.